

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (R,S) / S2 (PT) (R,S) Examination June 2023 (2019 Scheme)

Course Code: MAT206**Course Name: GRAPH THEORY**

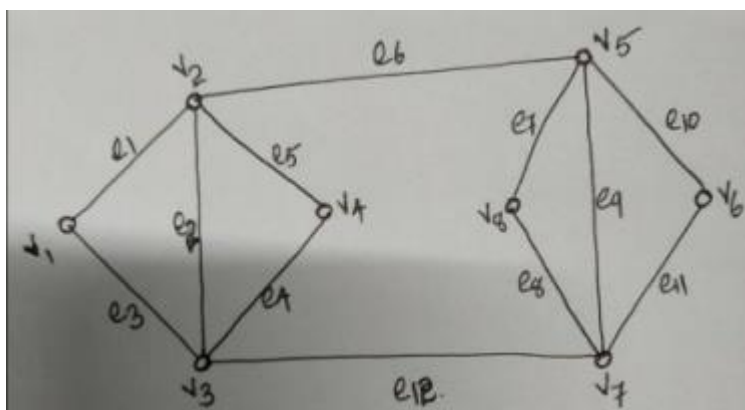
Max. Marks: 100

Duration: 3 Hours

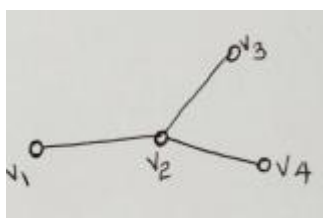
PART A*(Answer all questions; each question carries 3 marks)*

Marks

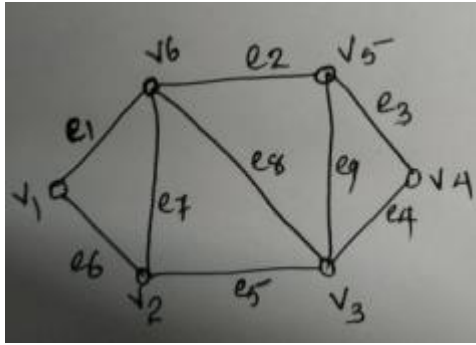
- | | | |
|---|---|---|
| 1 | Define Walk, Path and Circuit in a graph. | 3 |
| 2 | Define a connected graph with an example. | 3 |
| 3 | Check whether the following graph is Euler. If so find an Euler tour in it. | 3 |



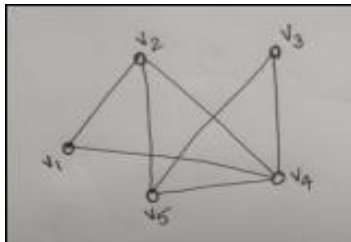
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|---|--|---|
| 4 | Define an equivalence digraph and give an example of an equivalence digraph on 4 vertices. | 3 |
| 5 | Define center of a graph. Find the center of the following graph | 3 |



- | | | |
|---|---|---|
| 6 | Prove that a binary tree on n vertices has $\frac{n+1}{2}$ pendant vertices. | 3 |
| 7 | List out any 5 different cut-sets and hence determine the edge connectivity of the following graph. | 3 |



- 8 Draw any two non planar graph with proper labelling. 3
- 9 Find the incidence matrix of the following graph. 3



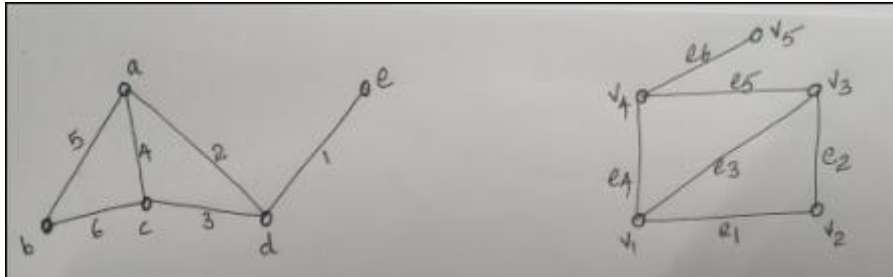
- 10 Define chromatic number with an example. 3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- 11 a) Prove that the number of odd vertices in any graph is always even. 7
- b) Check whether the following graphs are isomorphic or not 7



- 12 a) Prove that a simple graph with n vertices and k components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges. 8
- b) Define a sub graph, vertex disjoint sub graph and edge disjoint sub graph with an example. 6

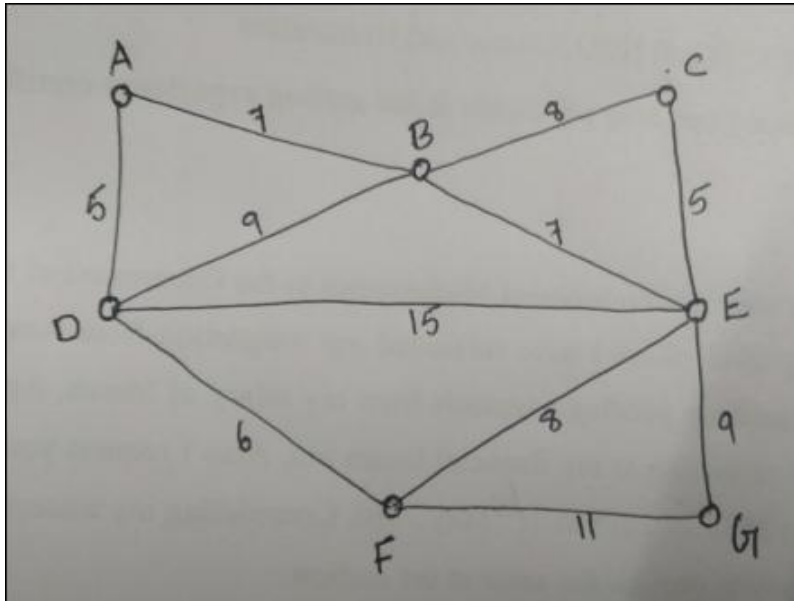
Module -2

- 13 a) Prove that a connected graph G is Euler if and only if degree of all the vertices in G are even. 8
- b) Distinguish between symmetric and asymmetric digraph with examples. 6

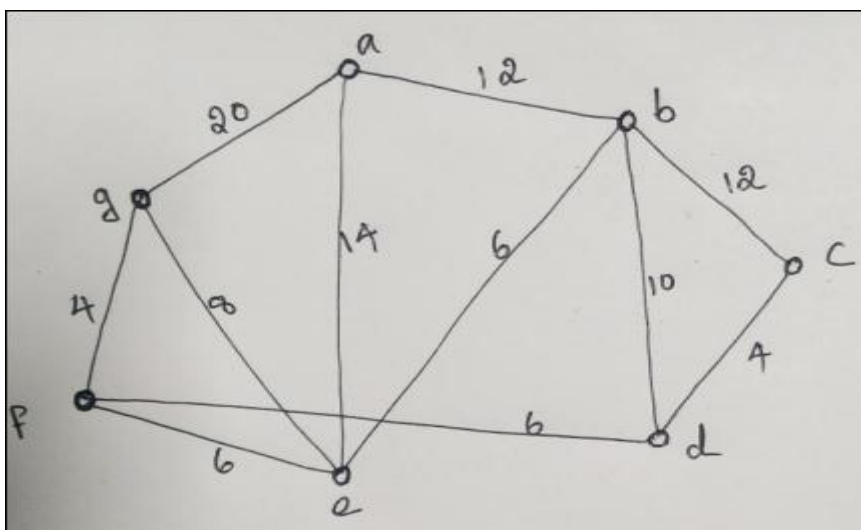
- 14 a) Explain travelling salesman problem. 7
- b) Prove that in a complete graph with n vertices there are $\frac{n-1}{2}$ edge disjoint Hamiltonian circuits, if n is an odd number ≥ 3 . 7

Module -3

- 15 a) Prove that a tree with n vertices has $n-1$ edges. 7
- b) Find the minimal spanning tree of the following weighted graph by using Prim's Algorithm 7



- 16 a) Prove that every connected graph has at least one spanning tree. 6
- b) Find the length of the shortest path from the vertex a to all other vertices of the given weighted graph G using Dijkstra's Algorithm 8



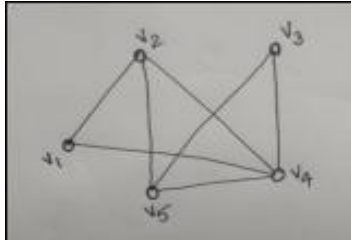
Module -4

- 17 a) State and prove Euler theorem on plane graphs. 8

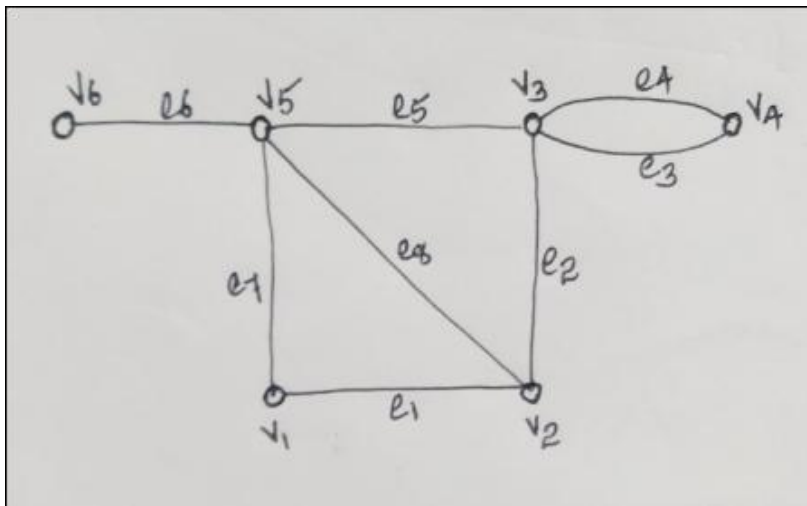
- b) Define vertex connectivity and edge connectivity of a graph with an example. 6
- 18 a) Prove that the maximum vertex connectivity of a connected graph G with n vertices and e edges is $\left\lfloor \frac{2e}{n} \right\rfloor$. 7
- b) Prove that every circuit has an even number of edges in common with any cut-set. 7

Module -5

- 19 a) Find the adjacency matrix corresponding to the graph given by 5



- b) Prove that every planer graph is 5 colorable. 9
- 20 a) Define a circuit matrix in a graph and hence find the circuit matrix of the following graph 6
- graph



- b) Prove that a graph with at least one edge is 2-chromatic if and only if it has no circuits of odd length. 8
